

VIJITH V T

AI Engineer | Generative AI & LLM Systems | Agentic AI | Backend Engineering | Enterprise Automation

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PROFESSIONAL SUMMARY

AI Engineer and Python Backend Developer with 10+ years of overall professional experience, including 3+ years building production-ready AI and automation systems, backed by an M.Sc. in Artificial Intelligence. Builds LLM-powered applications using FastAPI, LangChain, LangGraph, and RAG architectures, with hands-on experience across AI Agents, Vector Databases, and Model Context Protocol (MCP). Shipped real-world products including TalentHunt, an AI-powered recruitment platform that cut manual resume-screening effort by roughly 60-70% through LLM-driven contextual matching, an enterprise Knowledge Graph AI platform, and automation frameworks that increased test coverage by 30%+. Combines deep backend engineering (Python, FastAPI, Flask, PostgreSQL, Redis, Docker) with the ability to drive AI adoption across engineering teams, having mentored 200+ engineers through 50+ technical sessions on Generative AI, Python, and automation. Targeting AI Engineer, Generative AI Engineer, and Applied AI Engineer roles building production-grade Generative AI and Agentic AI systems.

CORE SKILLS

Generative AI & LLMs: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, Semantic Search, Conversational AI, LLM Integration, LLM APIs

Agentic AI: AI Agents, Multiagent Systems, Agentic AI Architectures, Multi-Step Reasoning, Tool Calling, Function Calling, Workflow Orchestration, Model Context Protocol (MCP)

AI Frameworks & Tools: LangChain, LangGraph, Vector Databases, Gradio

Backend Engineering: Python, FastAPI, Flask, Django, REST API Design, JWT Authentication, Role-Based Access Control (RBAC), System Integration, Asynchronous Processing (Celery), Software Architecture, Clean Architecture, Scalable Systems

Databases: Neo4j (Graph Database), PostgreSQL, MySQL, SQLite, Redis

Automation & Testing: Robot Framework, Playwright, Appium, Selenium, Automation Frameworks, SSH, RDP, Workflow Automation

Cloud, DevOps & Full-Stack: Docker, Git, Linux, CI/CD Fundamentals, Vercel, Supabase, React, TypeScript, Tailwind CSS

Engineering Enablement: Technical Mentoring, Curriculum Development, AI Enablement, Engineering Productivity

PROFESSIONAL EXPERIENCE

ThinkPalm Technologies Pvt. Ltd. — Associate Specialist, Academy & Research *Oct 2022 - Present*

- Design and implement **LLM-powered solutions and automation frameworks** for scalable enterprise applications, using FastAPI, LangChain, and RAG architectures.
- Architected an **Enterprise Knowledge Graph AI Platform** (Python, Flask, Neo4j, Celery, Redis, Docker) that centralized fragmented engineering knowledge from Git, Jira, commits, and documentation into a searchable graph, reducing manual effort to trace code ownership and dependencies.
- Architected a cross-VM **automation framework** using Robot Framework, Selenium, Playwright, and Appium, increasing automation coverage by **30%+** across Windows VM-based validation and testing workflows.
- Develop maintainable, Python-based backend services integrating AI capabilities with enterprise platforms, and prototype **Agentic AI proof-of-concepts** using LangGraph and Model Context Protocol (MCP) to evaluate tool-calling and multi-step reasoning patterns.
- Collaborate with cross-functional teams to enable intelligent workflows and ensure production readiness, focusing on performance and security.
- Mentor and train **200+ engineers**, delivering **50+ technical sessions** on Generative AI, Python, automation, and MLOps.

Kerala State Centre for Advanced Printing & Training — Guest Faculty, Computer Science *Sep 2018 - Oct 2022*

- Guided students in core computing concepts and emerging technologies, strengthening practical AI and programming skills.
- Developed and delivered interactive workshops connecting theoretical concepts to real-world AI and automation applications.
- Built student confidence through hands-on projects, fostering problem-solving skills and innovation.

Earlier Career *2015 - 2020*

Program Assistant, Central Adoption Resource Authority (2018-2020) | Project Assistant, National Health Mission (2017-2018) |

Commercial Apprentice, Kerala State Pollution Control Board (2015-2016)

Administrative and project-support roles prior to transitioning into technical education and software/AI engineering.

FEATURED AI PROJECTS

TalentHunt — AI-Powered Intelligent Recruitment Platform

Designed and developed an AI-powered Applicant Tracking System (ATS) that transforms traditional keyword-based recruitment into contextual, LLM-driven evaluation. Architected a modular FastAPI backend with MySQL, JWT authentication, and role-based access control. Built a Resume-JD Matching Engine using LLM + RAG pipelines with embeddings and vector retrieval, implementing multi-factor Fit Score logic with AI-generated explanations. Developed recruiter-controlled evaluation workflows and automated full application-lifecycle email communication. Reduced manual resume-screening effort by roughly 60-70% through intelligent contextual matching.

Tech Stack: Python, FastAPI, MySQL, JWT, LangChain, LLMs, RAG, Embeddings, Vector DB, React, REST APIs

Enterprise Knowledge Graph AI Platform

Architected and developed a backend platform centralizing fragmented engineering knowledge from Git, Jira, commits, and documentation into a searchable Neo4j graph. Built Python services and asynchronous processing pipelines using Celery and Redis, integrated multiple engineering data sources, and implemented AI-assisted repository analysis workflows. Reduced manual effort in tracing code ownership and dependencies, establishing a scalable foundation for AI-assisted engineering intelligence.

Tech Stack: Python, Flask, Neo4j, Celery, Redis, Docker

TechPulse AI Platform

Independently designed and built an automated content-intelligence platform for technology professionals. Implemented automated news aggregation, AI-powered summarization, blog-generation workflows, content categorization, and deployment. Reduced manual effort in monitoring technology news and demonstrated practical enterprise applications of Generative AI for content automation.

Tech Stack: Python, LLM APIs, Content Automation, Vercel

Sruthi Music Academy Management System – Full-Stack Learning Management Platform

Designed and developed a full-stack Student Management System supporting Students, Admins, and Institutions. Built a React + TypeScript frontend with role-protected dashboards, progress tracking, attendance, fee management, curriculum control, and trial-booking workflows. Implemented a Supabase backend with PostgreSQL, Row Level Security (RLS), authentication, storage, and Edge Functions for email automation and student lifecycle management. Integrated an AI-powered RAG chatbot for contextual assistance, and architected secure RBAC policies and a scalable database schema. Deployed web and Android builds via Capacitor.

Tech Stack: React, TypeScript, Supabase, PostgreSQL, RLS, Edge Functions, Tailwind CSS, TanStack Query, Capacitor, RAG

DigiLib – Modern Digital Library Management System

Designed and developed a web-based Digital Library Management System enabling users to browse, search, and manage digital book collections through a clean, responsive UI. Built using React, TypeScript, and Vite with Tailwind CSS and shadcn-ui for an accessible, scalable component architecture, focusing on maintainability, fast rendering, and organized content presentation.

Tech Stack: React, TypeScript, Vite, Tailwind CSS, shadcn-ui

AI Agent & MCP Experiments

Independently built multiple proof-of-concept agent implementations exploring orchestration, tool invocation, multi-step reasoning, LangGraph workflows, and Model Context Protocol (MCP) integration patterns. Strengthened applied expertise in Agentic AI architecture, directly informing current and future enterprise AI engineering initiatives.

Tech Stack: LangGraph, Model Context Protocol (MCP), Tool Calling, AI Agents, Multiagent Systems

EDUCATION

M.Sc. Artificial Intelligence — Dept. of Computer Science, University of Kerala, 2022

Master of Computer Applications (MCA) — IGNOU, 2018

B.Sc. Computer Science — College of Applied Science (IHRD), Dhanuvachapuram, University of Kerala, 2015

CERTIFICATIONS

Google IT Automation with Python Specialization | Google Data Analytics Specialization | Foundations of Project Management | Microsoft Excel: Beginner to Professional

ADDITIONAL EXPERIENCE

Founder, Sruthi Music Academy (Apr 2025 - Present) — Curriculum design, learning systems, and international online teaching.

Languages: English, Malayalam